THUNDERVOLTS

Reducing Household Electricity Overpayment in India

Presented by Team Abstract Minds | TechFusion 2.0

THE PROBLEM

Every Indian household tends to overpay electricity bills during summer months. High peak grid tariffs, inefficient usage patterns, and lack of real-time cost visibility create a significant financial burden across both urban and rural areas.

OUR SOLUTION

The Energy Optimizer App is an intelligent solution that helps Indian households optimize electricity consumption and reduce overpayment through smart, data-driven recommendations.

ML-POWERED FORECASTING

Leverages machine learning algorithms combined with public grid data and weather forecasts to predict optimal usage patterns and electricity costs in real-time.

ACTIONABLE RECOMMENDATIONS

Provides clear, simple guidance on what appliances to run and when, translating complex data into easy-to-follow steps that maximize savings.

INCLUSIVE DESIGN

Built for both urban and rural households across India, ensuring maximum social impact and accessibility for all communities regardless of technical expertise.

KEY FEATURES OF THE ENERGY OPTIMIZER

- Real-time Cost Monitoring
- Bill Prediction (XGBoost)
- Anomaly Detection
- Schedule Optimization
- Peak Hour Alerts
- AI Conversational Assistant
- Carbon Footprint Estimation
- Weather Integration

TECHNICAL ARCHITECTURE

Our Energy Optimizer app is built on a modern, scalable tech stack designed for real-time performance and seamless user experience across platforms.

Frontend Layer

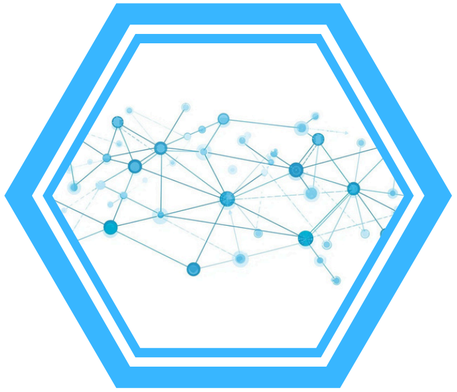
React/React Native

Backend Services

FastAPI Python

Database Layer

PostgreSQL + Redis



ML Integration

Forecasting Models

ML MODELS

Our Energy Optimizer app is powered by advanced machine learning algorithms that work together to analyze, predict, and optimize your household electricity consumption.

DEMAND FORECASTER

LSTM neural network predicts next-day grid demand on an hourly basis, enabling smart scheduling of appliances during low-cost periods.

BILL PREDICTOR

XGBoost regression model estimates your monthly electricity bill with high accuracy, helping you plan budgets and identify savings opportunities.

SCHEDULE OPTIMIZER

Greedy + Reinforcement Learning algorithms recommend optimal times to run appliances, maximizing cost savings while maintaining comfort.

DATA PIPELINE

Our robust data infrastructure ingests real-time information from multiple sources, ensuring accurate predictions and timely recommendations for optimal energy management.

GRID DATA INTEGRATION

Real-time grid data from POSOCO and state DISCOM tariff APIs provides accurate pricing signals and demand patterns for intelligent optimization.

WEATHER API INTEGRATION

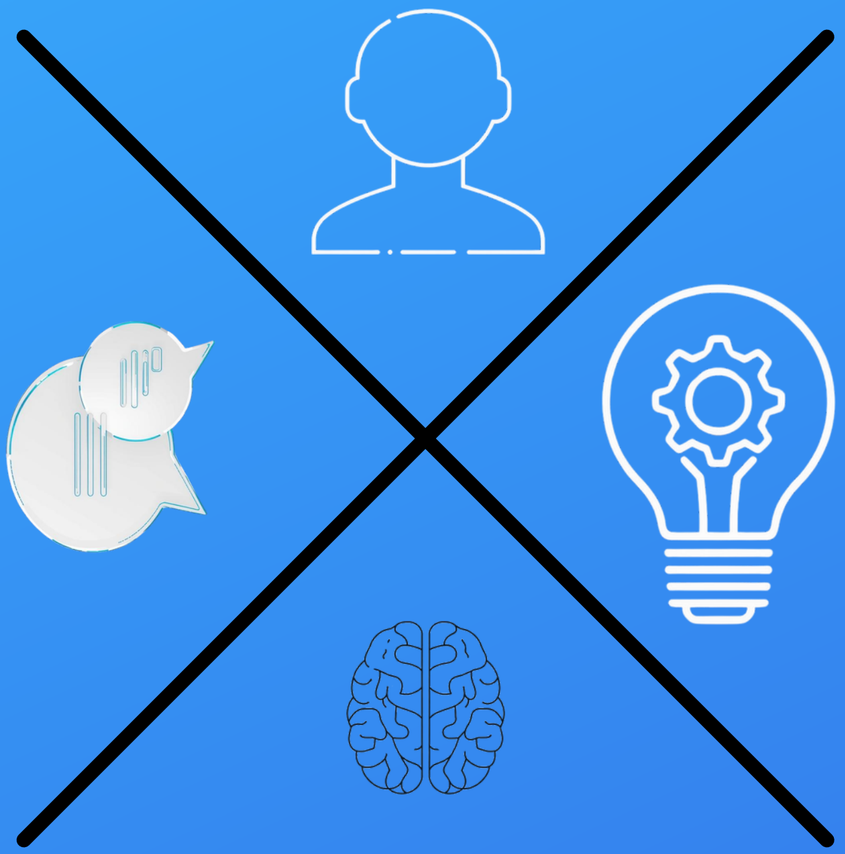
OpenWeatherMap API integration enables weather-aware demand forecasting, accounting for temperature and humidity impacts on consumption.

AUTO ML RETRAINING

Automated preprocessing, feature engineering, and model retraining pipelines ensure ML accuracy and robustness over time.

AI CONVERSATIONAL ASSISTANT LAYER

Powered by Claude Sonnet/Opus 4.6 LLM for intelligent reasoning and personalized energy advice. Provides bill explanations, anomaly insights, and scheduling tips with context-aware memory per user session.



IMPACT

Our Energy Optimizer app delivers meaningful social impact by addressing a universal pain point —electricity overpayment—while ensuring accessibility for all Indian households.

FINANCIAL RELIEF

Addresses a real, widespread financial burden affecting millions of Indian households. Empowers families to reduce monthly electricity bills through data-driven insights.

SOCIAL INCLUSION

Designed inclusively for both urban and rural populations, ensuring broad reach and accessibility. Simple interface makes advanced technology available to everyone.

ACTIONABLE INSIGHTS

Translates complex grid data into simple, actionable steps. Helps users make informed decisions to reduce costs and lower their carbon footprint simultaneously.

SCALABILITY & FUTURE VISION

Our roadmap extends beyond current capabilities, with ambitious plans to expand the Energy Optimizer into a comprehensive energy management ecosystem for Indian households.

SOLAR & EV INTEGRATION

Integrate rooftop solar panel monitoring and optimization. Add EV charging scheduling to leverage off-peak rates and maximize renewable energy utilization.

GRID DEMAND RESPONSE

Partner with DISCOMs for demand response programs. Enable users to participate in grid stability initiatives while earning incentives for load shifting.

NATIONWIDE ADOPTION

Expand AI capabilities for hyper-personalized energy saving plans. Vision to benefit millions of households across India with localized recommendations.

SUMMARY & NEXT STEPS

- Problem: Electricity Overpayment
- Solution: ML-Powered App
- Impact: Social Inclusion
- Environmental Benefits
- Next: Pilot Testing
- User Feedback Collection
- Scaling Development
- Nationwide Expansion

THANK YOU